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(1) Research Program/Project/Study Title: Frame-Based Expert Decision Support System on the Level of Risk Factors for COVID-19	
Proponent Name:	Cherry Joy A. Palomar
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Jelly L. Paredes, Rosel L. Dionio, Olga L. Llanera, Judith L. Vista, Leo Pula, Caster A. Kapunan, Nepthali B. Espartero, and Justin Kyle Capay
Campus/Satellite College/Address:	CAPSU Mambusao Satellite College
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation							
Municipality/City:							
Province:							
Region:							
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☒ Tech. Promotion/Commercialization</td> </tr> </table>	Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☒ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
Research:	Development:						
☐ Basic	☐ Pilot Testing						
☐ Applied	☒ Tech. Promotion/Commercialization						

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(7) Sector/Commodity

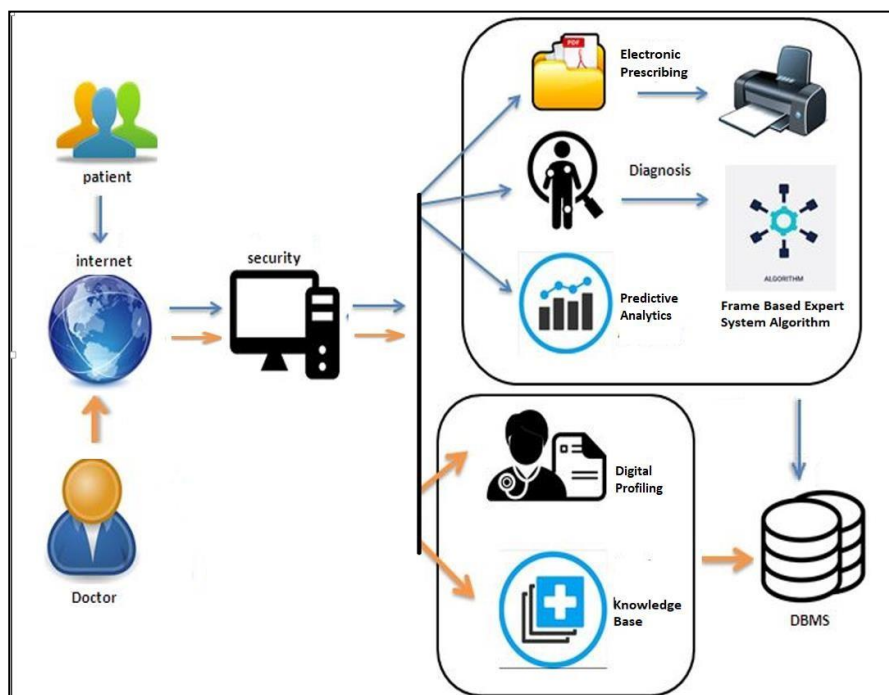
(8) Discipline/Field of Specialization

Computer Science

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The COVID-19 pandemic has led to a dramatic loss of human life worldwide and presents an unprecedented challenge to public health, food systems and the world of work. The economic and social disruption caused by the pandemic is devastating. The coronavirus disease 2019 (COVID-19) pandemic has caused increasing challenges for healthcare professionals globally. Although social distancing is the most effective way to contain the outspread of this virus, this is not easy to implement for healthcare professionals who require direct contact with COVID-19 patients and puts them under a high risk of being infected themselves. Frontline healthcare professionals are particularly vulnerable during this pandemic owing to their commitment to contain the disease. People are afraid in going to hospital for a fear that they may contract the virus, because hospitals are full of COVID-19 patients. Hospital is not a place people desire to go if they're at risk for obtaining an infection because there are other infected people there. Another reason people are afraid in seeing the doctor immediately, is that because of swab test. They are afraid of positive result that leads them to quarantine for 14 days. To help facilitate that all-important line of communication between doctor and patient, a Decision Support System was proposed to help the patients check their risks factor for COVID-19 through online platform. During this global pandemic, online platforms is an effective and sustainable solution for precaution, prevention and treatment to stem the spread of COVID-19. This proposed system offers clinical services when distance is a critical factor. Although this tool does not replace a medical examination, during Covid-19 pandemic, it reduces the spread of infection and avoids the need for a patient's visit. It delivers health-related services through the use of technology and informatics especially as a means of reducing the risk of cross-contamination caused by close contact.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)



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(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Statement of the Problem

- 1) There is no available technology that could provide risk factors of COVID-19 based on the user-defined symptoms.
- 2) Fear of patients in going to the clinic/hospitals because of COVID-19 virus.
- 3) There is no application that could give recommendation based on the risk factors of the patient.

Objectives

- 1) To design an interface that could determine the possible risk factors of COVID-19 based on the user-defined symptoms.
- 2) To develop a Decision Support System that allows the user to check their risk factors on COVID-19 through online platform.
- 3) To create a system that could generate recommendation based on the risk factors of COVID-19.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of – the-art of current technology / information from which the project proposal will take-off).

Using Clinical Ontologies to Build Knowledge Based Clinical Decision Support System for Novel Coronavirus (Covid-19) With the Adoption of Teleconferencing for the Primary Health Centres/satellite Clinics of Royal Oman Police in Sultanate of Oman

Inventors: P C Sherimon , Sherimon Vinu , Mathew Renchi , M Kumar Sandeep , V Nair Rahul , Shaikh Khalid , Al Ghafri Hilal Khalid , L Shuaily Huda Salim A,
Published: Apr 29, 2021

Abstract

Abstract A clinical decision support system (CDSS) expert system that incorporates artificial intelligence. Initially, the CDSS conducts the diagnosis to detect the symptoms of corona virus. We have used ontologies to represent the clinical protocol available to combat COVID-19. Ontologies include concepts (related to medical knowledge), attributes and semantic relationships between these concepts. Ontologies helps to represent the standard medical terms accurately, allows efficient knowledge sharing and reuse, and supports automatic reasoning. The CDSS considers and evaluates the medical history and other medical issues/complications of the suspected patient. So, once the symptoms of corona virus are confirmed in a patient, the CDSS will provide the necessary recommendations/suggestions/medications based on the patient's current medical condition. Further follow-up and guidance can be provided by the tele-services integrated in the system. Through tele services, the expert doctors can examine the patients and prescribe further tests/ medications in the follow-up session. e-Covid Architecture Covid Care Centre/Hospital'. Expert Doctor Lab Technician Patient Electronic Health Nurse Records Ontology Doctor Cncl symptoms Medical Application Admin Server Daaae Super Admin Analys1iReprting Diagnosis/ EMR/ Xray reports/ Lab reports etc Patient/User Figure: E-Covid Architecture 2

Depression Symptoms Prediction Model for Covid-19 From the Data of People Isolated and Home Quarantined

Inventors: Subramanian Appavu (Balamurugan) , Subramanian Senthamarai Kannan
Published: Jan 6, 2022

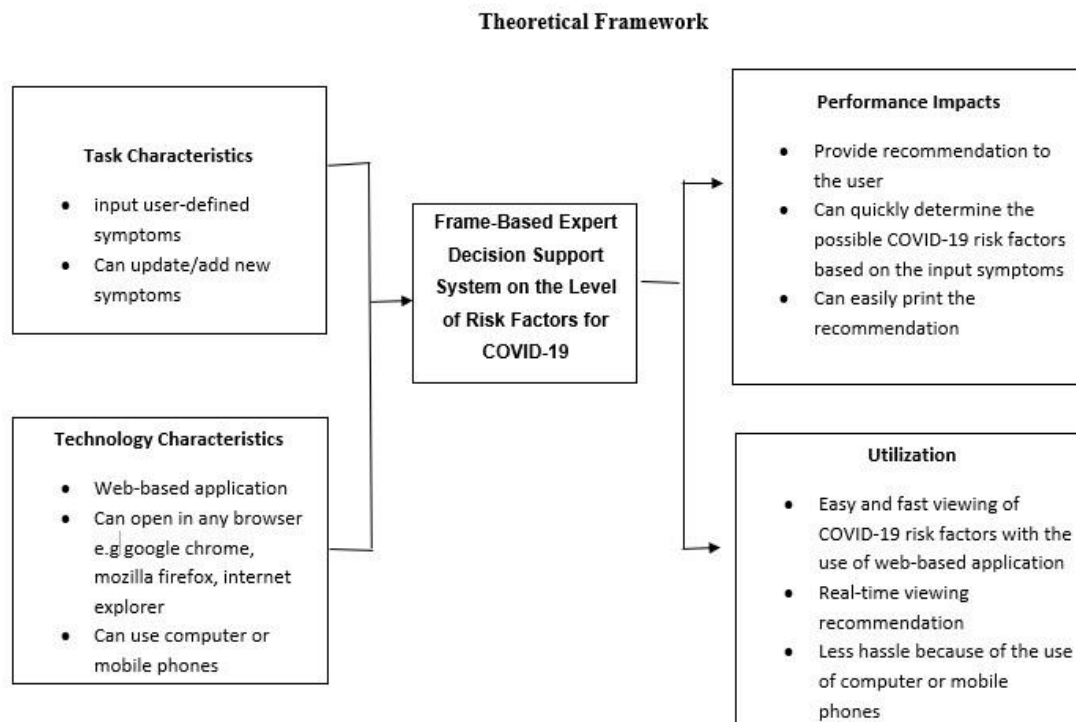
Abstract

DEPRESSION SYMPTOMS PREDICTION MODEL FOR COVID-19 FROM THE DATA OF PEOPLE ISOLATED AND HOME QUARANTINED OF THE INVENTION People affected by COVID-19 Isolation and Lockdown are twice as likely to experience clinically significant depressive symptoms as the general population. Using a rich data of people isolated due to COVID-19 collected through various social medias, this study has combined the logistic regression and Decision tree induction algorithm to develop prediction models to forecast comorbid depression for COVID-19 Isolated and Lock downed people. A benchmark model is developed using the depression symptom scale Patient Health Questionnaire (PHQ)-9 score at the study baseline as the predictor of PHQ-9 score 6 months later. The proposed ensemble model by combining the logistic regression and Decision tree induction algorithm, which included additional predictors, is compared to the benchmark model in both overall prediction accuracy and the accuracy of predicting major/severe depression. Adding predictors to the benchmark model is resulted in relatively small improvement in overall prediction accuracy, but it significantly

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improves the rate of true positive for predicting major/severe depression. The proposed prediction model will be the good starting points for the development of clinically useful predictive analytics tools that can assist in improving care for COVID-19 Isolated patents chronically with comorbid depression. Page I of 1 S. Appavu alias Balamurugan and S.Senthamarai Kannan The architecture of the proposed Depression Symptoms Prediction Model for Covid-19 from the Data of People Isolated and Home Quarantined is shown in Fig 1 below. Training and Validating of Data Collection Data Cleaning, the Proposed Ensemble model (Logistic Regression Decision Tree) using Cross Validation Supply test sample to predict depression Trained Model Predicting the Severity of Depression Fig 1: Architecture of the Proposed System for Predicting Depression Symptoms from the Data of People Isolated and Home Quarantined Various good practices for physicalhealth is highlighted in the Table 1. Table 1. Various Good Practices for Physical Health S.No Practices for Physical Steps to Reduce stress He alth trigge rs Connect with others 1 Enough sleep Keep your regular routine Make connections 2 Regular physical activity Limit exposure to news media Do something for others 3 Eat healthy Stay busy Support a family member or friend 4 Avoid tobacco, alcohol and Focus on positive thoughts drugs 5 Limit screen time Moral compass or spiritual life for support 6 Relax and recharge Set priorities

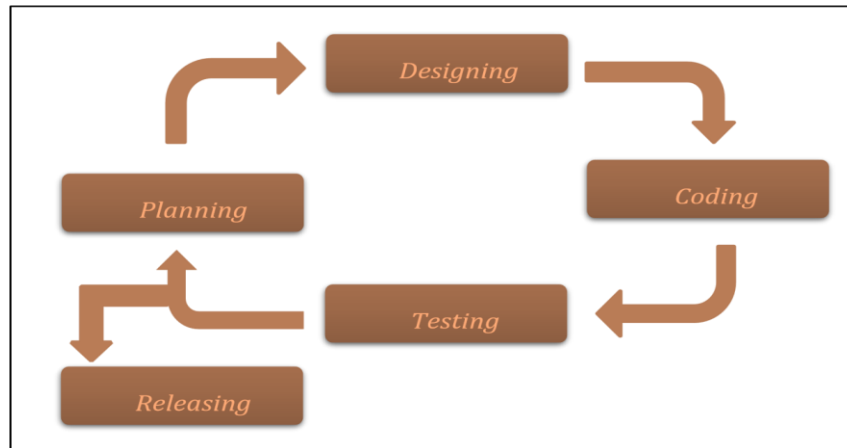
(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.



(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

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Extreme Programming



Extreme Programming (XP) is an agile software development framework that aims to produce higher quality software, and higher quality of life for the development team. XP is the most specific of the agile frameworks regarding appropriate engineering practices for software development (Alliance, 2018).

The XP Model is used for the development of medical diagnosis website applications. The project development requires rapid adaptation to changes that occur during the development of the application. Software development should be well planned, so that the software obtained quality and in accordance with the needs of users. The successful development of this software with XP can avoid the existence of patchwork in application development. Step by step research methods for medical record software are Planning, design, coding, testing and releasing. The XP methodology shows that every step always involves users, programmers, and parts of the test so that the applications created can be completed in a timely manner and in accordance with the needs of users. (I Gusti Ngurah Suryantara & Johanes Fernandes Andry, June 2018)

In this study, Extreme programming was used as XP provides a list of simple, specific, and seemingly raw principles and values that guide the software development process throughout the main four phases.

Extreme Programming focuses on deploying software modules piece by piece. Each module delivered to the client is checked and is then returned for error correction. It then undergoes several iteration processes so as to be delivered again. After the first module is sent, it is correspondingly followed by the following module to support the features of the first module in order to cover broader objectives regarding the project. With each module passing, the initial acceptance test is then gathered and formed as a whole to accomplish the objectives of the system.

Planning

During the 1st Iteration of the planning phase, all necessary information were gathered through an interview with 2 physicians. Possible problems are enumerated to identify the objectives. This gives knowledge to the research to conceptualize the design to be made once the objectives are determined. Identifying the necessary data that are used in determining a diagnosis of the patient is the most essential data that was acquired during the interview with the physician. The researcher assessed the results of the initial interview to be able to determine the feasibility of the study if developing a web based application would be probable.

Brainstorming and assessing the data acquired from the initial interview, the researcher is now able to come up with the probable objectives for the study developed.

Designing

Looking at the results of the planning phase, visualized system process where the system will be a web based application where two types of user can access the system. The administrator, will be the one will be responsible in managing the system. The patient, who is the second user of the system, will have the same feature of the system except in managing and maintaining it. Both users can encode the necessary data and go through the decision process.

Coding

In the coding phase, the identifying the tool to be used in done on this phase. Choosing the ReactJS as the programming platform was selected as the researcher is familiar with the environment to be used. Since the system is a web based application, the firestore database will be used as the cloud storage for the data to be saved on.

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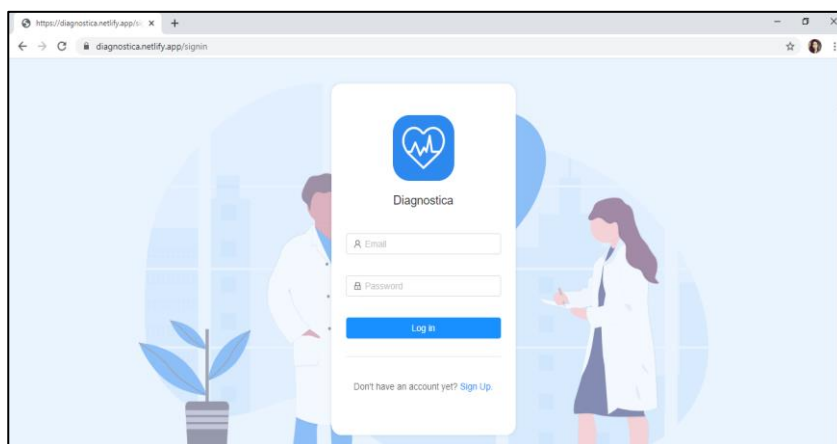
Testing

The type of unit testing to be used in the system to be developed is identified. Unit testing is used in conducting the testing of the system developed. Since coding has not yet initiated, testing would commence once coding has started.

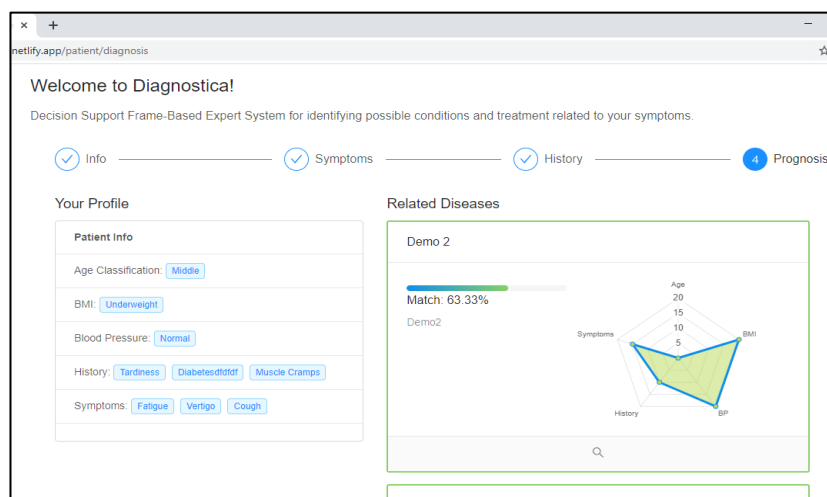
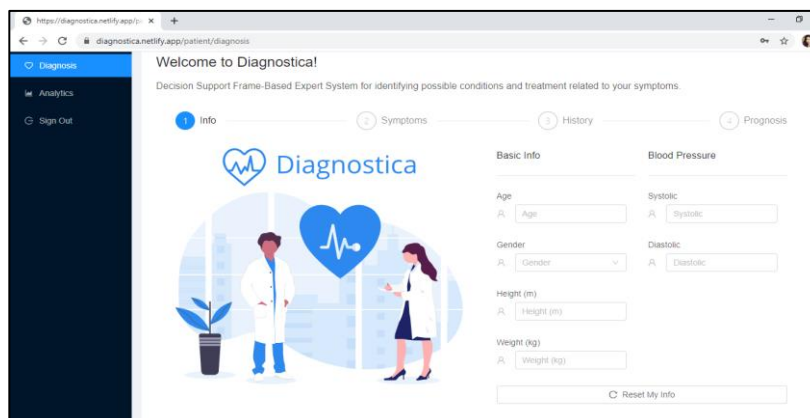
(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1												Year 2											
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Planning																								
2. Designing																								
3. Coding																								
4. Testing																								

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)



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(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The target user of this study is anyone who has the signs and symptoms of COVID-19. But for this proposed study the students of CAPSU Mambusao Satellite College will serve as the user of the system. The school nurse or school physician will be the administrator of the system, they are the one who will update the system.

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(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Hardware Development Cost

Web Server System Unit	Php 57,500.00
Keyboard and Mouse	Php 750.00
Printer	Php 4,750.00
UPS	Php 8,500.00
Total Cost	Php 71,500.00

System Development Cost

System Analyst	Php 50,000.00
Programmer	Php 40,000.00
GUI Designer	Php 22,000.00
Researcher	Php 32,000.00
Total Cost	Php 146,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)

CHERRY JOY ARROYO PALOMAR

Master of Science in Computer Science Graduate



PERSONAL DETAILS

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 Email Address: cherryjoypalomar@gmail.com
 Gender: Female
 Birth date: June 23, 1993
 Religion: Baptist
 Nationality: Filipino

PROFESSIONAL EXPERIENCE

August 2021 – Present **College Instructor**
 Computer Studies Department
 Capiz State University (Mambusao Campus)
 Poblacion, Mambusao, Capiz, Philippines

June 2013 – July 2021 **College Instructor**
 College of Computer Studies
 Filamer Christian University
 Roxas City, Philippines

EDUCATION

June 2014 – January 2021 **Central Philippine University**
 Master of Science in Computer Science
 Jaro, Iloilo City

June 2009 – March 2013 **West Visayas State University**
 Bachelor of Science in Information Systems
 Cum Laude

June 2005 – March 2009 **Bungsuán National High School**
 Valedictorian
 Bungsuán, Dumarao, Capiz Philippines

June 1999 – March 2005 **Bungsuán Elementary School**
 Valedictorian
 Bungsuán, Dumarao, Capiz Philippines